



ARUP DAS

Third-Year B.Tech Student — Computer Science & Engineering (AI/ML)

Kolkata, West Bengal, India | +91 7439399239 | dasarup0804@gmail.com

Portfolio: arup-portfolio-seven.vercel.app | LinkedIn: Arup Das | GitHub: arupdas0825

PROFILE

Date of Birth: 08/11/2004

Nationality: Indian

TECHNICAL SKILLS

Programming Languages

- Python
- JavaScript / TypeScript
- SQL

Frontend

- React, Next.js
- Vite, Tailwind CSS

Backend & APIs

- Node.js (REST APIs)
- FastAPI (Python)

Cloud & Data

- Firebase (Auth, Firestore)
- MySQL

Applied AI

- XGBoost / Scikit-learn
- Local LLM (Ollama)
- Prompt engineering
- Data analysis

Tools

- Git / GitHub
- VS Code, MS Office

CERTIFICATIONS

AI Fluency: Framework & Foundations

Anthropic, 2026

Claude 101

Anthropic, 2026

LEADERSHIP & ACTIVITIES

Class Representative

B.Tech CSE (AIML), Brainware University, 2024–25

Participant, “The Prompters” — TEXIBITION 2K26

Brainware University Tech Club, 2026

LANGUAGES

Bengali — Native (C2)

English

Listening B2 · Reading B2
Speaking B1/A2 · Writing B1
(IELTS preparation in progress)

Hindi

Listening B2 · Reading A2
Speaking B1/B2 · Writing A1

ABOUT

Third-year B.Tech CSE (AI/ML) student building intelligent, production-ready systems at the intersection of AI, software, and data. Passionate about solving real-world problems through machine learning, scalable applications, and continuous research-driven learning, with plans to pursue an M.Sc. abroad in AI, Data Science, or Computer Science.

EDUCATION

B. Tech, Computer Science & Engineering (AI/ML)

Aug 2024 – Present (June 2028)

Brainware University, Kolkata, India

CGPA: 8.0/10.0 (3rd year). Core coursework: AI, Machine Learning Algorithms, Data Structures & Algorithms, Probability & Statistics for Data Analytics, DBMS, Operating Systems, Computer Architecture, Discrete Mathematics, OOP.

Higher Secondary Education (10+2, Science)

Sept 2020 – June 2022

Uttar Dum Dum Vidyapith For Boys, Kolkata, India

Physics, Chemistry, Mathematics, Biology (Optional). Final grade: A+ (WBCHSE).

Secondary Education

Jan 2019 – June 2020

Uttar Dum Dum Vidyapith For Boys, Kolkata, India

Final grade: A (WBBSE, Madhyamik Pariksha).

KEY PROJECTS

Studytra — AI-Powered Study Abroad Platform

Personal Project, 2026

- Replaced the traditional study-abroad consultancy model with a production React SPA covering visas, costs, loans, and university timelines across 5 countries — built while navigating my own move to Austria.
- Integrated a Groq-powered (Llama 3.3 70B) AI advisor behind a secure serverless proxy, grounded in a hand-built country knowledge base — real decision support, not a demo chatbot.
- Shipped Firebase Auth, Firestore-backed community reviews, PDF country roadmaps via Python ReportLab, and a Cost Planner with live currency conversion.

Tech: React 18, Vite, Firebase, Groq API (Llama 3.3 70B), Supabase, Vercel Serverless Functions

Live: <https://studytra.vercel.app>

AtmosBridge — Federated AI Climate Intelligence Platform

Hack2Skill × Google Cloud, 2026

- Hackathon submission (Track: Clean Air & Climate Resilience) — fuses multimodal citizen reports, live OpenAQ sensor data, and satellite proxy signals to surface hyperlocal pollution events that city-wide monitors miss entirely.

- Engineered a physics-grounded XGBoost spike predictor (trained on 2,500 meteorological observations) delivering 6h/12h/24h PM2.5 forecasts with explainable feature importance — the first real ML model I've trained from scratch.
- Built a responsible-AI architecture: Gemini multimodal analysis is strictly grounded in verified telemetry (no hallucinated sensor values), with full data-provenance tagging and a human-in-the-loop authority triage system.

Tech: React, FastAPI, XGBoost, Google Gemini API, OpenAQ, Open-Meteo, Python, Docker, Tailwind CSS

Live: <https://atmosbridgeai.vercel.app>

German

A1 (all skills)

(Currently enrolled in A1–A2 course)

INTERESTS

Photography — landscape, street & architectural composition.

EverBond Wealth — Couple Financial Planning Platform *Personal Project, 2025–*

Present

- Three adaptive life-stage modes (Single → Committed Partner → Family Dynasty) that restructure the entire dashboard and calculation engine in real time.
- Financial engine replicates a multi-sheet Excel investment model: cascading budget allocations, multi-asset splits (equity, debt, commodities, crypto), and risk-tiered sub-allocation — all computed client-side with Excel-accurate precision.
- QR-code partner pairing via Firebase Auth & Firestore real-time sync, packaged as an offline-capable PWA.

Tech: React 19, Vite, Firebase (Auth, Firestore), Zustand, Recharts, Framer Motion

Live: <https://everbond-wealth.vercel.app>

CodeFolio — Developer Intelligence & Networking Platform *Personal Project, 2026*

- Converts a developer's public GitHub activity into a quantitative 0–1000 Developer Score via a custom multi-factor algorithm: commit velocity, code-health metrics, and a Shannon-entropy technical-diversity measure.
- Evolved from a scoring tool into a developer social platform — DevFeed, a Connections hub, and an AI Intelligence Suite for resume analysis — demonstrating how focused tools can grow into applied-AI products.
- GitHub OAuth (Firebase Auth), REST + GraphQL API integration, zero-config demo mode, and real-time interactive data visualizations.

Tech: Next.js 16, React 19, TypeScript, Tailwind CSS, Firebase, GitHub REST/GraphQL API, Recharts, Framer Motion

Live: <https://codefolio-dev.vercel.app/>

NeuralCode — Privacy-First AI Code Translator & Explainer *Personal Project, 2025*

- Translates, explains, and documents code across 14 languages using a 100% local LLM (CodeLlama via Ollama) — zero cloud calls, zero source-code exposure.
- Temperature-stratified prompt framework: deterministic prompts for translation, structured prompts for explanation (incl. “Explain Like I’m 10” mode), zero-temperature for language classification — my first deep dive into precise LLM behavior control.
- Four async API endpoints (translate, explain, comment, detect) with input validation, timeout handling, and a companion technical report reviewing Codex, CodeT5, and CodeLlama architectures.

Tech: Next.js 14, TypeScript, Tailwind CSS, Ollama (local LLM inference), Prompt Engineering